LONDON METROPOLITAN UNIVERSITY

1st sit Coursework - 01 Question Paper **Spring Semester 2026**

Module Code:	**CS5054NT**
Module Title:	**Advanced Programming and Technologies**
Module Leader:	**Mr. Sujan Subedi/Mr. Sandip Dhakal**

Coursework Type:	Group
Coursework Weight:	This coursework accounts for 50% of your total module grades.
Submission Date:	First Milestone: May 4, 2026 Final Submission: May 21, 2026
Coursework is given out:	Week 5
Submission Instructions:	Submit the following to college's MST portal before 02:00 PM on the due date: ● A report (document) in .pdf format in the MST portal or through any medium the module leader specifies.
Warning:	London Metropolitan University and Itahari International College take plagiarism seriously. Offenders will be dealt with sternly.

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PLAGIARISM

You are reminded that there exist regulations concerning plagiarism. Extracts from these regulations are printed overleaf. Please sign below to say that you have read and understand these extracts:

Extracts from University Regulations on Cheating, Plagiarism and Collusion

Section 2.3: “The following broad types of offence can be identified and are provided as indicative examples ….

Cheating: including taking unauthorised material into an examination; consulting unauthorised material outside the examination hall during the examination; obtaining an unseen examination paper in advance of the examination; copying from another examinee; using an unauthorised calculator during the examination or storing unauthorised material in the memory of a programmable calculator which is taken into the examination; copying coursework.

Falsifying data in experimental results.

Personation, where a substitute takes an examination or test on behalf of the candidate. Both candidate and substitute may be guilty of an offence under these Regulations.

Bribery or attempted bribery of a person thought to have some influence on the candidate’s assessment.

Collusion to present joint work as the work solely of one individual.

Plagiarism, where the work or ideas of another are presented as the candidate’s own.

Other conduct calculated to secure an advantage on assessment.

Assisting in any of the above.

Some notes on what this means for students:

Copying another student's work is an offence, whether from a copy on paper or from a computer file, and in whatever form the intellectual property being copied takes, including text, mathematical notation, and computer programs.

Taking extracts from published sources *without attribution* is an offence. To quote ideas, sometimes using extracts, is generally to be encouraged. Quoting ideas is achieved by stating an author's argument and attributing it, perhaps by quoting, immediately in the text, his or her name and year of publication, e.g. “$e = mc^2$ (Einstein 1905)". A *reference* section at the end of your work should then list all such references in alphabetical order of authors' surnames. (There are variations on this referencing system which your tutors may prefer you to use.) If you wish to quote a paragraph or so from published work then indent the quotation on both left and right margins, using an italic font where practicable, and introduce the quotation with an attribution.

School of Computing, FLSC

CONTRACT CHEATING

Contract cheating (also known as assessment outsourcing, commissioning or ghost writing) is when someone seeks out another party, or AI generator service, to produce work or buy an essay or assignment, either already written or specifically written for them or the assignment to submit as their own piece of work.

Contract cheating undermines the integrity of the academic process and devalues the qualifications awarded by the university. Students are reminded that academic integrity is a fundamental principle of our institution. Engaging in contract cheating not only impacts the individual's academic record but also the reputation of the university.

Students are encouraged to seek support if they are struggling with their coursework. The university offers a range of resources, including academic counseling, tutoring services, and workshops on study skills and time management. Utilizing these resources can help students achieve their academic goals without resorting to dishonest practices.

Penalty:

- Failure in the Module: The student must re-register for the same module, and the re-registered module will be capped at a bare pass.
- Ineligibility to Continue on the Course: Where re-registration of the same module, or a suitable alternative, is not permissible, the student will not be able to continue on the course. Additionally, the following penalty will be applied to the student's final award:
    - Undergraduate Honors: The student's final classification will be reduced by one level.
    - Unclassified Bachelors: Downgraded to Diploma in Higher Education.
    - Foundation Degree: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Certificate in Higher Education.
    - Masters: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Postgraduate Diploma.

Reporting and Consequences:

Instances of contract cheating will be thoroughly investigated, and students found guilty will face the penalties outlined above. It is the responsibility of every student to ensure that their work is their own and to avoid situations that could lead to accusations of academic misconduct.

By adhering to these standards, students contribute to a fair and equitable academic environment, ensuring the value and recognition of their qualifications are maintained.

Deliverables

You are required to submit two components before the submission deadline.

1. Dynamic System:
   - The **project**, including the generated **SQL schema**, must be uploaded to a **GitHub** repository.
2. Report in PDF format.
   - Provided report structure and guidelines in **Task B** should be included.
   - Additionally, in your cover page include **a link to the GitHub repository** in your report, ensuring that the repository is publicly accessible.

Task A: System Development

Please note that the scenario described below cannot be used as your coursework project. You are required to propose a new and original project idea that is distinct from any other projects from students and does not align with the provided project scenario.

Project Scenario

For example, if we propose a **Library Management System (LMS)** named **"CollLib",** as our coursework project, the system would serve as a comprehensive web-based platform to **manage library related activities** within an educational institution. This project would allow students to explore key aspects of web development, including **user authentication, role-based authorization, data management, CRUD operations, and user interface design.**

The detailed information on system components is explained below:

1. **Database Design and Development** **[5 marks]**
   a. The database design will include a normalized data model capturing all essential entities and relationships, such as student records, courses, and user accounts, ensuring data integrity and minimizing redundancy.
   b. Relationships between entities will be mapped using foreign keys and constraints, and the database will be optimized for performance with indexing and transaction management to ensure efficient and consistent data operations.
   c. Backup and recovery mechanisms will be implemented to protect against data loss or corruption, ensuring the system's reliability and resilience.
2. **Admin Dashboard:** **[10 marks]**
   a. The admin dashboard would enable administrators to manage book records by **creating, updating, deleting, and viewing** detailed information such as book title, ISBN number, genre, published date and author. The admin also approves the user registration request.
   b. Admin can issue books to the users with proper issue and return date recorded.
   c. Additionally, it would facilitate book management by allowing administrators to assign books, manage rack number, and organize books in racks.
   d. The system would also **generate reports** on analysis of books' issues vs availability, favorite books by number of issues, genre to aid in decision-making and planning.
   e. Furthermore, user management capabilities would allow the approval and management of user accounts with appropriate roles and permissions.
3. **User Portal:** **[10 marks]**
   a. The user portal would provide users with a platform to **register and manage their profiles**, allowing them to **register**, **log in, view, and update** their personal information, including contact details and passwords.
   b. They can register with personal data (e.g., name, date of birth, contact details) and academic data (e.g., courses enrolled, level, year). Their registration request to be approved by admin.
   c. Users would also have access to their **records**, issued books, return due, number of

books issued, the most issued books. The portal would enable users to access genre-related materials.

d. The system will include a search feature that allows users to find books information, via author and ISBN, using a **search bar**. This functionality will enable users to quickly access relevant details by entering keywords or criteria related to their search.

e. The system will include an **"Apply" feature** that allows users to request book available and they are interested in. This functionality will enable users to book material directly through the application.

f. User can wish-list multiple books, for which session can be used.

4. Authentication & Authorization: **[5 marks]**

a. The system would employ **secure login protocols** to ensure that only authorized users can access the platform.

b. Access control should be **role-based**, distinguishing between **administrative users and normal users** i.e., enforcing restrictions on functionalities based on the user's role, and proper encryption program.

c. **Session** can be widely used to complete the **authentication and authorization process**.

d. Redirect Management (Filter)

i. The system would efficiently **manage URLs**, directing users to the appropriate pages based on their actions and roles. Unauthorized access attempts would result in redirects to the login page or an error page, while session management would ensure that users remain logged in across different pages until they choose to log out or their session expires.

5. MVC Architecture: **[5 marks]**

a. The system will be developed using the **Model-View-Controller (MVC)** architecture, which separates the application into three interconnected components.

i. The **Model** component to manage the data of student records and course information.

ii. The **View** component, built with JSP and CSS, will be responsible for presenting data to users, tailored to their roles (e.g., admin or student).

iii. The **Controller** will process user requests, coordinate with the Model and Service to retrieve or update data and determine the appropriate View to display.

b. In addition to the MVC components, the system will utilize packages such as **util and service,** along with other necessary packages, to enhance functionality and maintainability.

i. The util package will include utility classes and methods for handling common tasks like validation, date formatting, and other reusable functions.

ii. The service package will encapsulate business logic, facilitating interactions between the Controller and the data access layers.

6. Frontend Design and Development: [5 marks]

a. The user interface would be built using **JavaServer Pages (JSP)** for dynamic content and **CSS** for styling, ensuring a **responsive** and user-friendly design that accommodates a range of devices, from desktops to mobile phones. The use of frameworks like Bootstrap and others is prohibited. **You can use CSS media query and Flexbox properties for responsive design.**

b. JavaScript should be only used as needed for **enhancing the design and user experience** of the application. The core functionality and server-side logic will be managed by the Java-based components of the system.

7. **Additional Features:** **[5 marks]**

   The system can improve with the inclusion of:

   a. **About page** to provide information about the institution.

   b. **Contact page** offering support details and a form for inquiries and so on.

Note:

While this scenario serves as a detailed example of a web application project, students are not permitted to use it for their coursework. Instead, they must propose and develop a unique project centered around an ethical topic, ensuring that each project is distinct. These projects should incorporate similar technical components, such as MVC architecture, role-based access control, and responsive design, providing students with practical experience in web development using Java, JSP, CSS, J2EE and MySQL. The coursework aims to enhance students' skills in system design, implementation, and problem-solving, while also considering ethical implications.

Other Program Requirements: **[5 marks]**

1. **Validation and Exception Handling:**
   - During the data registration process, it is imperative to implement validation checks to verify the existence of an account. This can be done by cross-referencing a unique identifier, such as a phone number, to ensure that the account does not already exist. ISBN for a book should be unique.
   - Proper exception-handling mechanisms must be in place to manage potential errors gracefully in the entire system. Must have error pages implemented.
2. **Programming Styles:**
   - The Java code should follow standard naming conventions and include meaningful comments to enhance code readability and maintainability. These practices are crucial for maintaining clarity in the code and ensuring that it adheres to professional standards.
3. **Proper Use of Classes and Methods:**
   - Object-oriented programming principles should be applied efficiently to reduce code redundancy. This involves utilizing methods and classes effectively to create a clean and modular codebase, emphasizing reusability and maintainability.
4. **Use of Proper Messages:**
   - When creating a new account, the system must provide appropriate prompts if any entered information is invalid. For example, if a user enters numerical data in the full name field, the system should alert the user with a clear and specific error message, guiding them to correct the input.

Task B: Report (50 marks)

Word Count: 16,000 (word count starts from introduction till conclusion.)

1. **Introduction**

   This section should include a clear and concise introduction to the project, covering the following elements:

   - **Title**: Provide a descriptive title for your project.
   - **Purpose**: Explain the purpose of the project and why it is being developed.
   - **Audience**: Identify the intended audience for your application.
   - **Aims and Objectives**: Outline the main aims and objectives that the project seeks to achieve.
   - **List of Features**: Identify the features to implement in your system and outline them in one line.

2. **Wireframe [8 marks]**

   - Present the wireframes of your application alongside your prototype designs, showing the layout and structure of the user interface. This should include all major screens, such as the login page, dashboard, and any other key pages.

3. **Java Classes Explanation [8 marks]**

   - Provide a concise explanation of the Java classes used in your application. This section should include:

     i. **Class Diagram**: A visual representation of the classes, their relationships, and interactions.

     ii. **Methods Explanation**: For each Java class, list the methods used and explain their purpose and functionality.

4. **Test Cases [10 marks]**

   - When documenting your test cases, ensure that each test case is thoroughly detailed with a comprehensive test table and clear steps to achieve the expected result. This section should demonstrate the full range of tests conducted to validate the functionality and robustness of your project.

5. **Coursework Development and Critical Analysis [10 marks]**
   Document the development process of your coursework according to the team member, including the tools used to complete the system. For each tool, provide:

   - **Tool Description**: What the tool is and its primary purpose.
   - **Usage**: How the tool was used in the development process.
   - **Evidence**: Include screenshots or other evidence demonstrating the use of the tool.
   - You can take **Database Tables and ER Diagrams** as an example to start with.
     - **List of Database Tables:**
       Identify and list the tables utilized within your application, such as Students, Courses, and Enrollments, each serving distinct roles in data management.
     - **Entity-Relationship (ER) Diagram:**
       Provide a comprehensive ER diagram that illustrates the relationships between these tables, highlighting primary and foreign keys.
     - **Tool Utilized:**
       Specify the tool employed, such as XAMPP, and its primary function in designing and managing the database schema.
     - **Application of the Tool:**
       Detail how the tool was used to develop a complete database and generate the ER diagram, ensuring the integrity and efficiency of the data structure.
     - **Evidence:**
       Include relevant evidence, such as screenshots of the database tables and the ER diagram, to substantiate your implementation process.

   - **Critical Analysis [6 marks]**
     Conduct a critical analysis of the system, evaluating its performance and identifying any challenges or difficulties encountered during development. This section should include:
     - **System Breakdown**: Discuss any issues, errors, or obstacles faced during development, supported by evidence such as screenshots.
     - **Evaluation**: Provide your opinion on the overall effectiveness of the system.

6. **Conclusion**

   Summarize the key findings from your coursework. Reflect on what you have learned, any challenges faced, and how you overcame them. Consider the overall success of the project and any areas for future improvement.

7. **References (Bibliography)**

   List all the sources referenced throughout your report in a properly formatted bibliography.

   Note that report styles and structure should be effectively implemented. [8 marks]

Important Details:

1. **Mandatory Use of Technologies:**

   Failure to incorporate required technologies such as Java, Java EE, MySQL, or JSP may result in an automatic failure of the coursework.

2. **Program Functionality:**

   Ensure that your program operates correctly on at least two different PCs prior to submission to avoid technical issues.

3. **Viva Attendance:**

   Non-attendance at the viva examination will lead to a failure in the system evaluation.

Task C: Milestone

Milestone submission is one of the most critical parts of this coursework. Students are required to submit the coursework in milestone 2. Further details are provided below.

A. **Milestone 1: May 4**

   Part 1: The task will involve completing the wireframe design, prototype design, and UI/UX development using JSP and CSS by creating a new project for the coursework.

   Part 2: Completion of Database Normalization, ERD creation, and database creation.

   Part 3: The task will be to complete the user registration and login features. Proper authentication must be implemented for the login feature, including encryption, file handling, session management, filter, and cookies. Additionally, all validation cases should be handled with appropriate messages displayed to the users.

   Part 4: The task will involve completing the features including the CRUD operations for at least the admin dashboard ensuring proper validations and messages are displayed throughout. User features can be completed in the final part of the work. If completed, will be reviewed.

   Part 5: Draft of the report for review.

   Note that you should include each session mentioned in the milestone including the code.

   Note that inconsistency in milestone submission may lead to failure

B. **Final: May 21**

   Submit all the tasks as per report requirements: Final project zip and report in MST. Github link of the project will be included in the cover letter which is a public repo of your work.